Anatomy and Taxonomy of Work Systems (M1.5)

Woojin Park
Life Enhancing Technology Laboratory
SNU Department of Industrial Engineering

Anatomy and Taxonomy of Work Systems

- Understanding work systems requires two complementary lenses:
  - Anatomy — What every work system contains
  - Taxonomy — How systems differ structurally

Anatomy of Work Systems

- Every work system contains:
  - Purpose / value proposition
  - Outputs (products, services, decisions, knowledge, experiences)
  - Customers / stakeholders
  - Processes and task structure
  - Participants / roles (human and/or machine)
  - Technology and tools
  - Information and rules
  - Environment and infrastructure
- Performance is an emergent property of interaction among these elements.
- This anatomy applies to factories, hospitals, AI systems, digital platforms, and even games.

Example: Smartphone Game as a Work System

- Work system analysis:
  - Purpose → Entertainment, engagement, challenge
  - Outputs → Experience, progression, ranking
  - Participants → Player + software agent
  - Processes → Gameplay loops, reward systems
  - Technology → Smartphone + game engine
  - Information → Rules, scores, feedback
  - Environment → Physical + digital interface

Why Taxonomy Matters

- A useful taxonomy distinguishes work systems in ways that guide analysis and design.
  It helps reveal meaningful differences in:
  - Coordination structure
  - Information flow
  - Variability patterns
  - Performance metrics
  - Suitable improvement methods

Why Taxonomy Matters

- Three complementary taxonomies characterize work systems along different dimensions:
  - A three-axis structure-based taxonomy
  - A two-axis complexity-based taxonomy
  - An operation-based work system taxonomy (Professor Jinwoo Park)
- Each taxonomy reveals a different facet of work systems.

Structure-based Taxonomy

- A **three-axis framework** that classifies systems along:
  - Volume–variety structure
    - Unit / Small-Batch Production
    - Mass Production
    - Continuous Process Production
  - Automation and agency
    - Manual Systems
    - Mechanized Systems
    - Automated Systems
    - Agentic (AI-Augmented) Systems
  - Temporality
    - Continuous / Repetitive Systems
    - Project-Based Systems

Focus: *How the system is structurally organized*

Structure-based Taxonomy

- Axis 1: Volume–Variety Structure
  - Unit / Small-Batch Production (Woodward, 1965)
    - Low volume, high customization
    - Skilled labor, flexible routing
    - Decentralized decision-making
    - High variability
    - Examples: construction, custom equipment
    - Improvement focus → flexibility, expertise coordination

Structure-based Taxonomy

- Axis 1: Volume–Variety Structure
  - Mass Production
    - High volume, low variety
    - Sequential workflow
    - Standardized tasks
    - Mechanization
    - Examples: automotive assembly
    - Improvement focus → throughput, variability reduction

Structure-based Taxonomy

- Axis 1: Volume–Variety Structure
  - Continuous Process Production
    - Extremely high volume
    - Continuous material flow
    - Heavy capital investment
    - Centralized control
    - Examples: oil refining, power plants
    - Improvement focus → stability, uptime, control systems

Structure-based Taxonomy

- Axis 2: Automation Level and Agency
  - Manual Systems
    - Humans sense, decide, act
    - Tools assist only
    - Human-centered control loops
    - Safety mainly ergonomic

Structure-based Taxonomy

- Axis 2: Automation Level and Agency
  - Mechanized Systems
    - Machines amplify effort
    - Humans retain decision authority
    - Task–tool alignment critical

Structure-based Taxonomy

- Axis 2: Automation Level and Agency
  - Automated Systems
    - Machines execute control logic
    - Humans supervise
    - Monitoring replaces execution
    - Interface design critical

Structure-based Taxonomy

- Axis 2: Automation Level and Agency
  - Agentic (AI-Augmented) Systems
    - Systems adapt or generate
    - Authority partially distributed
    - Governance and accountability central
    - Opaque reasoning risks

Structure-based Taxonomy

- Axis 3: Temporality
  - Continuous / Repetitive Systems
    - Ongoing operation
    - Stable architecture
    - Flow optimization
    - Examples: manufacturing, hospitals
    - Tools → Lean, statistical process control (SPC), capacity and flow balancing

Structure-based Taxonomy

- Axis 3: Temporality
  - Project-Based Systems
    - Temporary organization
    - Unique deliverable
    - Defined start and end
    - Milestone-driven coordination
    - Examples: construction, software development
    - Tools → PERT/CPM, risk management, agile methods

Structure-based Taxonomy

- Classification Examples:
  - Automotive assembly → High-volume, automated, repetitive
  - Custom engineering → Low-volume, human-driven, project-based
  - Digital banking → High-volume, automated/agentic, ongoing service
  - Smartphone game platform → High-volume, automated, ongoing service (episodic engagement)

Complexity-based Taxonomy

- A **two-axis framework** that classifies systems along:
  - Interaction complexity
    - Low to high
  - Work domain complexity
    - Low to high

  Focus: *How cognitively and operationally complex the system is*

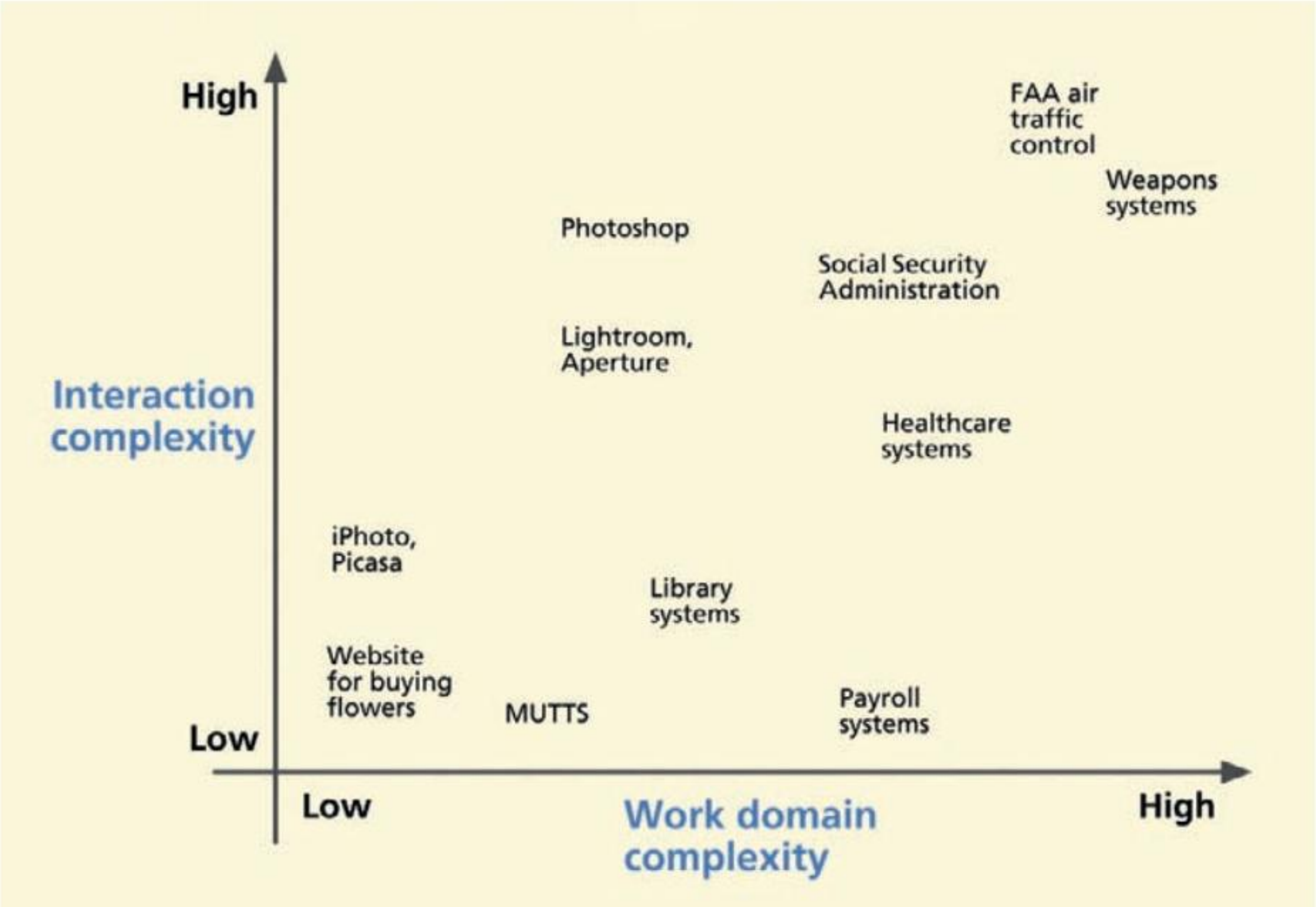
Complexity-based Taxonomy

- Axis 1: Interaction Complexity
  - How demanding the interaction is for the user.
  - Includes:
    - Number of interaction steps
    - Cognitive load
    - Decision density
    - Error consequences
    - Multitasking requirements
    - Information integration demands
  - High interaction complexity → Requires strong interface design and feedback structure.

Complexity-based Taxonomy

- Axis 2: Work Domain Complexity
  - How complex the underlying work domain is.
  - Includes:
    - Technical knowledge required
    - Regulatory constraints
    - Interdependencies
    - Organizational coordination
    - Risk and uncertainty
  - High domain complexity → Requires deep domain modeling and decision support.

Complexity-based Taxonomy



Operation-based Taxonomy

- Classifies work systems based on **dominant operational characteristics**
- Focuses on **how work is actually performed in real-world settings**
- Complements:
  - Structure-based taxonomy (system configuration)
  - Complexity-based taxonomy (cognitive & domain difficulty)
- Particularly useful for:
  - Industrial engineering analysis
  - Productivity improvement
  - Work measurement and system design
- Bridges **theory and practice** in work system classification

Operation-based Taxonomy

- **Simple Work Systems**
  - Tasks require **minimal prior training or expertise**
  - High **repeatability** and low variability
  - Easily analyzed using:
    - Time study
    - Motion study
  - Human labor is the primary resource
  - Examples:
    - Manual material handling
    - Basic assembly tasks
    - Repetitive service tasks

Operation-based Taxonomy

- **Machine-Based Work Systems**
  - Work involves **operation of powered equipment**
  - Clear distinction between:
    - Human time
    - Machine time
  - Performance depends on:
    - Human–machine coordination
    - Interface usability
  - Key issues:
    - Utilization
    - Idle time
    - Operator workload
  - Examples:
    - CNC machine operation
    - Excavator driving
    - Data entry systems

Operation-based Taxonomy

- **Flow Production Systems**
  - Organized as **sequential and interdependent processes**
  - High **throughput and standardization**
  - Requires:
    - Line balancing
    - Synchronization of tasks
  - Strong coupling between workstations
  - Vulnerable to bottlenecks
  - Examples:
    - Assembly lines
    - Mass manufacturing systems

Operation-based Taxonomy

- **Batch Production Systems**
  - Work is processed in **groups or batches**
  - Balances:
    - Efficiency (scale)
    - Flexibility (variety)
  - Key challenges:
    - Setup time
    - Scheduling
    - Queue management
  - Not fully continuous, but not purely discrete either
  - Examples:
    - Passenger transport by aircraft
    - Freight transportation (per truckload)
    - Book publishing
    - Live entertainment (theater/auditorium)
    - Laundry services

Operation-based Taxonomy

- **Project-Based Work Systems**
  - Produces **unique, one-off outputs**
  - Defined by:
    - Clear start and end
    - Milestones and deadlines
  - High uncertainty and coordination complexity
  - Requires:
    - Planning tools (e.g., Gantt, PERT/CPM)
  - Often involves multidisciplinary teams
  - Examples:
    - Shipbuilding
    - Aircraft development
    - Large-scale engineering projects

Operation-based Taxonomy

- **Logistics Work Systems**
  - Focus on **movement of materials and information**
  - Integrates:
    - Transportation
    - Inventory
    - Information systems
  - Strong dependence on:
    - Network coordination
    - Timing and synchronization
  - Increasingly data-driven and automated
  - Examples:
    - Supply chain systems
    - Distribution centers
    - Warehouse operations

Operation-based Taxonomy

- **Knowledge and Service Work Systems**
  - Based on **expert knowledge and human judgment**
  - High variability and low predictability
  - Output quality is often:
    - Difficult to measure objectively
  - Requires:
    - Communication
    - Decision-making
    - Customer interaction
  - Examples:
    - Healthcare (doctors)
    - Legal services
    - Education
    - Software development

Integrating Work System Taxonomies

- Work systems can be understood through **three complementary taxonomies**
  - **Structure-based taxonomy**: How is the system organized?
  - **Complexity-based taxonomy**: How difficult is the system to operate?
  - **Operation-based taxonomy**: What kind of work is being performed?
- These taxonomies are **not competing, but mutually reinforcing**
- Each taxonomy reveals a **different facet of the same work system**
- A complete understanding of a work system requires **all three perspectives**

Mini-Summary

- Anatomy defines universal components
- Taxonomy differentiates structural types
- These lenses enable principled work system redesign across industrial, service, digital, and AI domains